Nguyễn Quốc Tuấn

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Introduction

AI Engineer - Computer Vision

AI Engineer with 1+ year of production experience building end-to-end Computer Vision systems, from model training to real-time edge deployment. Focused on turning slow, manual, or unsafe workflows into fast, automated processes. Experienced on fusing/quantizing model for edge deployment.

Experience

Vinsmart Future — Ho Chi Minh City
AI Engineer — *Aug 2026 – Sep 2026*

Truck tarp Violation Detection | Python, DETR, ResNet

- Built a full violation detection pipeline to flag uncovered construction-material trucks.
- Achieved 91% end-to-end accuracy and 97% recall in automated testing on private datasets.

Household Re-ID system for Smart-home | Python, ViT, PyTorch

- Developed a system assigning IDs to residents, neighbors, and pets to flag unknown persons or animals, finetuning a ViT backbone with a custom retrieval head.
- Achieved 95% mAP and 99% Rank-1 in automated testing on private datasets.

Rainscales Viet Nam — Ho Chi Minh City
AI Engineer — *Jun 2025 – Jun 2026*

Warehouse Inventory Counting System| Python, ResNet, YOLO, OCR, decord, OpenCV

- AI camera pipeline replacing manual pallet counting, improved worker safety and 5x faster.
- 90–94% video-to-crop accuracy, 81% counting accuracy (measured on onsite production video).

Event Analytics Pipeline | Python, NVIDIA DeepStream, ServiceMaker

- Deployed real-time video analytics on Jetson, handling multiple RTSP streams with a store-and-process pipeline (MP4/JSON chunking, rule engine, rolling buffering), publishing events to Kafka.
- Scaled to 8 cameras at 25fps on Orin Nano and 60 at 25fps (fully loaded) on AGX Orin; also deployed on Jetson Thor for a 5-camera site.

Model Research & PoC | Python, YOLO, ResNet, TensorRT

- Trained a warehouse object detector with 3M images, 16 classes, achieving 85% mAP@0.5.
- Trained a PPE compliance classifier (helmet/vest/coverall), achieving 88% accuracy, false alarm 5%.
- Fused and quantized models into lightweight TensorRT engines for real-time inference on Jetson.

Education

Ho Chi Minh City University of Technology (HCMUT – VNUHCM) — Ho Chi Minh City
Bachelor of Science – Computer Science — *2026*

Research

SMC2L: Self-Supervised Mixup Contrastive Continual Learning | Thesis | Paper

- Graduation thesis proposing a self-supervised continual learning method (contrastive loss + pixel mixup + distillation) to reduce **catastrophic forgetting**.
- 92% accuracy on satellite dataset; accepted at IAAA 2026 (publish on Springer LNNS, SCOPUS). Authors: Tin Duc Duong, **Tuan Quoc Nguyen**, Trang Hong Le.

SAM3-DeepStream | Python, C++, SAM3, TensorRT, NVIDIA DeepStream

- Personal research adapting Meta's SAM3 into DeepStream's nvinfer, exporting a TensorRT engine with a custom C++ mask/bbox parser, supporting on-the-fly model updates.
- Reached 10 FPS per stream on 4-camera RTSP 1080p (RTX 5070, 4GB).

Skills

Coding: Python, C/C++, SQL (MySQL, SSMS)
Machine Learning: PyTorch, OpenCV, YOLO, ResNet, ViT, OCR
Edge: TensorRT, Triton, NVIDIA DeepStream, NVIDIA Jetson (Orin, Thor)
MLOps: Kafka, Docker, Linux (Ubuntu)

Certifications

IELTS Band 6.5
HackerRank SQL (Advanced)